Carbonatite-hosted REE prospectivity under extreme label scarcity in Canada: national-scale sampling and transferability, and a semi-supervised deviation network in three geological provinces

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Abstract

Carbonatite-hosted rare earth element (REE) prospectivity models must learn from a few tens of occurrences and no confirmed barren ground. A national-scale model of Canadian carbonatite-hosted REE deposits reported test AUCs above 0.9 with S2-grid label augmentation, negatives drawn from the ground least similar to the occurrences and random splits. Rebuilt on our national data, this design gives a test AUC of 1.000; with one label per occurrence, random background and a realistic prevalence of 0.3%, the same random forest reaches an AUC of 0.903 and an average precision (AP) of 0.292. Models also do not transfer between geological provinces: a province's model ranks another province's occurrences at or below chance in most cases, and the provinces are fully separable in the evidence layers. We therefore model provinces separately, with the Superior Province (15 occurrences) as the focus and the Cordilleran Orogen and Grenville Province as contrasts, and compare a semi-supervised Deviation Network (DevNet), which treats unexplored ground as unlabelled, with random forest (RF), XGBoost, tuned tree ensembles and two positive–unlabelled learners under random and 50–200 km spatially blocked cross-validation. In Superior DevNet reaches an AP of 0.45 under random cross-validation and 0.42–0.25 under blocks, against 0.27–0.28 and 0.20–0.12 for RF and XGBoost; with 15 occurrences its margin is consistent across repeats but not statistically conclusive. One mapped alkalic-intrusion class carries over a third of DevNet's blocked AP. In Cordillera and Grenville no model transfers between blocks, and synthetic positives do not compensate for label scarcity.

Keywords: Mineral prospectivity mapping, Rare earth elements, Carbonatite, Deviation network, Positive–unlabelled learning, Spatial cross-validation, Transferability, Data leakage

1 Introduction

Rare earth elements (REEs) are critical to low-carbon energy, electronics and defence technologies, yet economic concentrations are geologically uncommon (Wang, Fan, Zhou, Yang, & She, 2020). Most REE resources are hosted by carbonatites and associated alkaline intrusions, whose distinctive but partly overlapping geophysical and geochemical signatures make them natural targets for data-driven mineral prospectivity mapping (MPM; Simandl & Paradis, 2018; Verplanck, Van Gosen, Seal II, & McCafferty, 2014). In Canada, however, confirmed carbonatite-hosted REE occurrences are few and spatially clustered (Fig. 2): after near-duplicates are merged, a single geological province holds between about ten and thirty of them.

Two properties of this problem undermine standard supervised classification. First, positives are extremely scarce: mineralisation is a rare event (Xiong & Zuo, 2018), known occurrences are only a partial and biased sample of the mineral system (Coolbaugh, Raines, & Zehner, 2007; Parsa & Carranza, 2021; Zuo et al., 2021), and under k-fold cross-validation each fold holds only a handful of them, so any estimate is dominated by which occurrences happen to be held out. Second, there are no true negatives. Ground without a recorded occurrence is unexplored rather than barren, and designating it negative is an arbitrary modelling choice whose outcome depends on which background points are drawn (Zuo & Wang, 2020). Together these properties motivate positive–unlabelled (PU) and semi-supervised anomaly-detection formulations, in which mineralised sites are treated as rare anomalies relative to a large, mostly barren but unlabelled population (Bekker & Davis, 2020; Y. Chen & Wu, 2019; Luo, Farahbakhsh, Hore, & Müller, 2026; Xiong & Zuo, 2016, 2021).

The Deviation Network (DevNet; Pang, Shen, & van den Hengel, 2019) is a semi-supervised deep anomaly detector that learns an anomaly score from a few labelled anomalies and a large unlabelled pool, pulling the scores of unlabelled data towards a Gaussian reference while pushing labelled anomalies a fixed margin above it. Luo et al. (2026) adapted it to MPM (DEEP-SEAM), encoding geology as distances to mapped units and adding neighbourhood statistics of the evidence layers. Whether such a model outperforms well-tuned discriminative baselines when positives number in the low tens has not been tested under evaluation designs that control for the specific ways in which small, clustered label sets inflate scores. The closest prior study, a national-scale model of Canadian carbonatite-hosted REE ± Nb deposits (Parsa et al., 2024), reported test AUCs of 0.91–0.96 with random forests and one-dimensional convolutional networks. Its labels were augmented by replacing each occupied S2 grid cell with its four child cells, its negatives were drawn from the ground least similar to the positives, and it was evaluated by stratified random 70/30 splits of the augmented labels. A follow-up study recommended synthetic labels generated by generative adversarial networks (GANs) for the same commodity (Parsa et al., 2025).

Each of these design choices can make the held-out test easier than prospecting new ground. A national model also answers a different question from a provincial one. Pooling every occurrence of a country raises the label count, but it mixes occurrences from geological provinces whose geophysical settings, and whose barren ground, differ completely; whether a model trained on such a pool predicts within any one province, and whether one province's occurrences say anything about another's, is rarely tested (Meyer & Pebesma, 2021; Ploton et al., 2020).

This study starts at the national scale. We rebuild the S2-grid design of Parsa et al. (2024) on national data, compare it with a design that keeps one label per occurrence, draws background at random and evaluates at a realistic prevalence, and test whether models transfer between geological provinces. Because they do not, the main part of the study models provinces separately: the Superior Province of the Canadian Shield, the province in which labels are scarcest relative to area, with the Cordilleran Orogen and Grenville Province modelled with the identical pipeline as contrasting cases. All analyses share one occurrence table and one input rule; inputs whose value depends on the extent of the study area are computed per geological province, so that a point has the same value nationally and provincially. We ask: (i) how much of the high AUC of the national S2-grid design comes from its sampling and augmentation, and do models transfer between provinces? (ii) within Superior, does DevNet outperform tree-based learners, including tuned ensembles and XGBoost, and positive–unlabelled learners once every model sees the same data and folds, and does any advantage survive spatially blocked validation? (iii) how much of the skill comes from geological map classes that mark known complexes? and (iv) can synthetic positives compensate for label scarcity? We close by interpreting the provincial models with SHapley Additive exPlanations (SHAP), using a rank-based rule for the direction of each feature's effect, and by checking the Superior maps against an untrained REE resource and the mapped Kapuskasing Structural Zone.

2 Geological setting and data

2.1 Carbonatite-hosted REE systems and their geophysical expression

Carbonatites occur predominantly in Precambrian cratonic areas, and individual provinces record repeated emplacement over hundreds of millions of years, implying long-lived, episodically reactivated trans-lithospheric pathways (Woolley & Bailey, 2012). A component of their REE endowment is attributed to subcontinental lithospheric mantle refertilised by REE- and CO_2-rich metasomatic fluids (Hou et al., 2023; Yaxley, Kjarsgaard, & Jaques, 2021). Each element of this source–transport–trap system has a mappable geophysical proxy (Fig. 1). Depth to the lithosphere–asthenosphere boundary (LAB) indexes the thickness and edge geometry of the cratonic root that hosts the fertile mantle source, which exerts first-order control on magmatic ore systems (Griffin, Begg, & O'Reilly, 2013; Groves & Santosh, 2021; Hoggard et al., 2020); Moho depth indexes crustal thickness; and proximity to faults and to the loci of maximum horizontal gravity and magnetic gradient ("worms"; Hornby, Boschetti, & Horowitz, 1999) proxies the structures that channel ascending melts. At

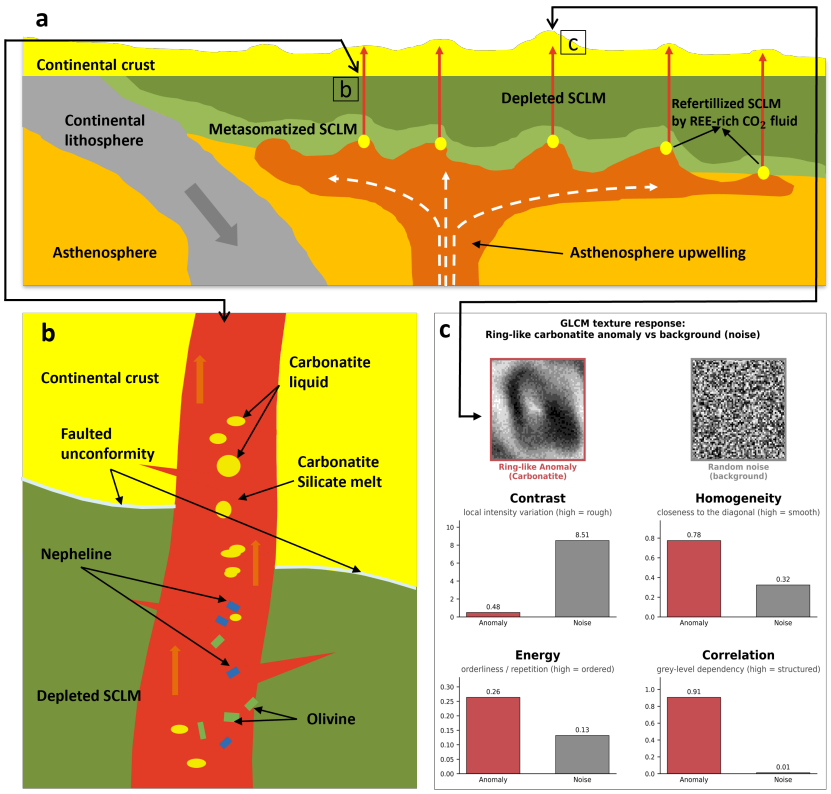


Fig. 1 Source–transport–trap model for carbonatite-hosted REE systems and its geophysical proxies. (a) Source: refertilised subcontinental lithospheric mantle near the LAB at the craton margin, indexed by LAB depth and margin proximity; panel modified after Hou et al. (2023). (b) Transport along trans-lithospheric faults, indexed by fault and worm proximity and Moho depth; modified after Yaxley et al. (2021). (c) Trap: GLCM texture response of a ring-like anomaly and of random noise.

the emplacement site, dense and magnetic alkaline–carbonatite complexes produce positive gravity anomalies and intense circular to annular magnetic anomalies (Drenth, 2014; Thomas, Ford, & Keating, 2016). Grey-level co-occurrence matrix (GLCM) textures summarise the local two-dimensional heterogeneity produced by such concentric patterns without assuming a target diameter (Cooper & Cowan, 2005; Hall-Beyer, 2017), unlike explicit radial-symmetry detectors tuned to ring-complex size (Aranha, Porwal, Sundaralingam, & González-Álvarez, 2021).

2.2 Study areas

The national study area is the union of the onshore geological provinces, orogens and platforms of the Geological Map of Canada (Map 1860A; Wheeler et al., 1996), without the continental shelves and oceanic crust (12.3 million km^2). The three provincial study areas are the Superior Province, the Grenville Province and the Cordilleran Orogen polygons of the same map. Carbonatite-hosted occurrences of the Phanerozoic,

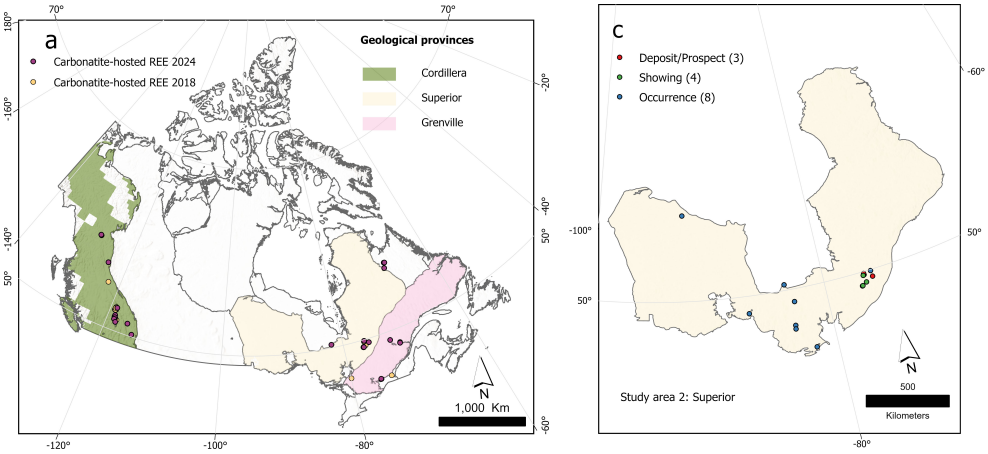


Fig. 2 (a) Carbonatite-hosted REE occurrences in Canada and the three modelled provinces; coloured areas are the sampleable domains. (b) The 15 deduplicated Superior occurrences used as positives, by development class.

accreted Cordillera are geophysically distinct from those of the Canadian Shield (Parsa et al., 2024), which suggests that a single cross-terrane model would not transfer; Section 4.1 tests this directly. Windows of one province enclosed by another and smaller than 2,000 km^2, such as the Churchill-basement windows of the Monashee and Malton complexes in the Cordillera, were assigned to the surrounding province.

The Superior Province is the world's largest Archean craton. Carbonatite–alkaline magmatism in the Canadian Shield clusters into episodes at about 2.68 Ga, 1.90–1.88 Ga and 1.16–1.08 Ga (Rukhlov & Bell, 2010), and in the Superior Province many of the complexes are aligned along crustal-scale structures, most prominently the Kapuskasing Structural Zone (KSZ; Gittins, Macintyre, & York, 1967; Percival & West, 1994). The 15 Superior occurrences used here (Fig. 2b) comprise a cluster around the Montviel and Lac Shortt carbonatites in the eastern Abitibi region of Québec (Deposit/Prospect and Showing class) and occurrences dispersed across the southern and western Superior Province, including Nemegosenda Lake, Lackner Lake, Kapuskasing, Martison Lake, Prairie Lake and Carb Lake. The Superior Province thus combines the fewest occurrences per unit area of the three provinces with a spatial structure, one tight cluster plus scattered singletons, in which a model can either learn the cluster or learn what the scattered occurrences have in common.

2.3 Occurrence labels

Positive labels were compiled from the carbonatite-hosted records of the USGS global REE occurrence database (Orris, Seo, Briggs, Dunlap, & Cocker, 2018) and the occurrence table of Parsa et al. (2024). Alkaline-hosted occurrences were excluded after a preliminary leave-block-out classification showed them to be geophysically separable from carbonatite-hosted ones. Records within 2 km of each other were merged by single linkage, so that drill holes or zones of one deposit, listed once or in both sources, count once; without this step near-identical points from one deposit fall into both training

and validation folds. Merging and the data-coverage filter (Section 2.6) reduced 21, 13 and 51 pooled records to 15 (Superior), 12 (Grenville) and 27 (Cordillera) positives. Each positive keeps the development class of Parsa et al. (2024): Deposit/Prospect, Showing or Occurrence (Superior: 3, 4 and 8). The national label table takes the three provincial tables as they are and adds the 13 occurrences elsewhere in Canada, merged by the same rule (five in the Churchill Province, three in the St. Lawrence Platform, two in the Hudson Bay Lowlands, two in parts of the Cordilleran Orogen outside the study polygon and one in the Slave Province); two records outside a study polygon but within 2 km of one of its occurrences were merged into that occurrence. After the data-coverage filter the national table holds 66 occurrences (Fig. 2a).

2.4 Geophysical and structural evidence layers

Table 1 lists the evidence layers. Seventeen national rasters cover the trap and the deep-architecture components: Bouguer gravity and its derivatives and filtered versions, total magnetic intensity and its differential reduction to the pole with derivatives (Jobin, Véronneau, & Miles, 2017; Miles & Oneschuk, 2016), together with long-wavelength gravity and magnetics, LAB and Moho depth from the Tri-National Critical Minerals Mapping Initiative release (McCafferty et al., 2023). Eleven of the gravity and magnetic grids are processed products supplied by Getech Group plc; four of these are gridded versions of the public Natural Resources Canada compilations and seven are standard derivatives of them (Table 1). Two line layers (ancient passive margins, D.C. Bradley, 2008; faults) and four point layers (shallow and deep gravity and magnetic worms; McCafferty et al., 2023) enter as normalised proximities. GLCM contrast, homogeneity, energy and correlation were computed in 9×9-pixel windows on the Bouguer anomaly, its 40 km high-pass, the reduced-to-pole magnetics and the total magnetic intensity. Following Luo et al. (2026), the mean and standard deviation of the same four grids within 10 km of each point were added as neighbourhood statistics. Radiometric grids (K, eU, eTh) were considered but not used, because their coverage of the study areas is incomplete and uneven; whole-rock geochemistry is too sparse to give a value at every map cell. The province polygons of Wheeler et al. (1996) define the study areas only; no distance-to-province-boundary feature was used.

2.5 Geology

Geology enters from a hexagonal H3 grid (resolution 7, cells of about 5 km^2) attributed from the Geological Map of Canada (Wheeler et al., 1996). Each cell carries a major lithology class (for example Igneous Intrusive Alkalic or Metamorphic Gneiss Paragneiss), its dominant era, and flags for cells crossed by a lithological or an age (period) contact; each point takes the attributes of the cell that contains it. The lithology class enters as one-hot columns (a class gets its own column if at least 10 of the region's 5,000 training-background points fall in it; rarer classes share one), together with the era one-hot and the two contact flags. Following Luo et al. (2026), geology also enters as distances (km) to six elements of a generalised map, from which every lithological unit smaller than 20 connected cells (about 100 km^2) and the alkalic-intrusion class

Table 1 Evidence layers, grouped by mineral-system component. Native resolution in brackets. All rasters were sampled at points in their native projection and resolution.

Component	Layer(s)	Source
Source / lithosphere	Depth to LAB (0.1°); Moho depth (0.25°)	McCafferty et al. (2023)
Deep architecture	Bouguer gravity upward-continued 30 km and its horizontal-gradient magnitude (HGM) (0.031°); long-wavelength (deep-source) RTP magnetics and its HGM (0.02°)	McCafferty et al. (2023)
Transport	Proximity to faults; to ancient passive margins; to shallow and deep gravity and magnetic worms	McCafferty et al. (2023); D.C. Bradley (2008)
Trap: gravity	Bouguer anomaly (BA), BA first vertical derivative, BA total horizontal derivative (THD) (2 km)	Jobin et al. (2017); gridded by Getech
	BA THD tilt, BA 40 km and 100 km high-pass (4th-order Butterworth) (2 km)	Getech derivatives
Trap: magnetics	Total magnetic intensity (TMI) (0.01°)	Miles and Oneschuk (2016); gridded by Getech
	Differential reduction to pole (DRTP), DRTP 1VD, DRTP THD, DRTP THD tilt (0.01°)	Getech derivatives
Anomaly form	GLCM contrast, homogeneity, energy, correlation (9×9 window) on BA, BA 40 km high-pass, DRTP, TMI; mean and standard deviation within 10 km of the same grids	this study
Geology	H3 cell (resolution 7): major lithology class, era, lithological- and age-contact flags; distances to contacts and units of a generalised map (Section 2.5)	Wheeler et al. (1996)

were removed: lithological contacts, age contacts, greenstone (extrusive) units, mafic–ultramafic intrusions, high-grade metamorphic units and sedimentary units. After pruning (Section 3.2) the inputs number 48 in Superior (24 geophysical and structural layers, 3 neighbourhood statistics, 2 regional distances, 15 lithology columns, 2 era columns and 2 contact flags), 55 in Cordillera and 58 in Grenville.

At the scale of a carbonatite complex a bedrock map can mark the target itself: in Superior, 6 of the 15 occurrences fall on cells mapped as alkalic intrusive rock, against 16 of the 3,000 evaluation-background points (0.5%). A mapped alkaline complex is legitimate evidence for an explorer, but a model can score well simply by looking it up. We therefore also report the models without the point-level map classes (the lithology, era and contact columns), keeping the geophysical and structural layers, neighbourhood statistics and regional distances (29, 31 and 33 inputs after pruning). This measures what the models achieve on ground where the map does not show a complex.

2.6 Sampleable domain and background

Rasters were clipped to each study-area outline buffered outward by one GLCM window radius (8 km), so that windows near the boundary read real data, and sampling and prediction were restricted to the domain where every layer holds data and every texture window is computable. This domain covers 98.6% of the Superior, 97.2% of the Grenville and 73.5% of the Cordilleran outline and 8.09 million km^2 (65.6%) of Canada; one Cordilleran occurrence (Lempriere) and one St. Lawrence Platform

occurrence (Francon Quarry) fall outside it and were dropped. Background candidates are the nodes of a 0.1° grid inside the domain and more than 10 km from any known occurrence (19,371 in Superior). From these, two disjoint pools were drawn once per region at random: an evaluation background of 3,000 points and a training background of 5,000 points. Every model in this study is trained on the positives of its training fold plus the same 5,000 training-background points and scored on the held-out positives plus the same evaluation background. The prevalence in the scored pool is therefore $15/3{,}015 \approx 0.5\%$ in Superior, which is the AP of a random ranking.

For the national analyses the same rules were applied to Canada as a whole. Inside the study provinces the national background points are provincial ones (in Cordillera and Grenville random subsets of them, so that the density matches Superior's), and elsewhere background was drawn at Superior's density, 41% of the candidate nodes, with evaluation and training background split 3:5 as in each province. The national table holds 66 occurrences, 20,855 evaluation-background and 34,761 training-background points (prevalence 0.32%).

3 Methods

Figure 3 summarises the workflow. Code for every analysis, including the exact numbers in every table, is available in the repository listed under Code availability.

3.1 National and provincial inputs

Three inputs depend on the extent of the area over which they are computed: raster values at the edge of a clip (a coarse pixel whose centre falls outside the buffered outline is nodata), the GLCM grey levels, which are quantised between the 2nd and 98th percentiles of the clipped raster, and the generalised geology map, which is built from the H3 cells in the outline's bounding box plus a margin. All three were computed per geological province, with the provincial rasters and bounding boxes for the study provinces, so that every national value equals the provincial one at the same point. Feature pruning (Section 3.2) and the H3 class list are applied to the table on which a model is trained: national models use the 65 inputs pruned on the national table, and the model of a province uses that province's inputs and network capacity, also when it is applied to another province.

3.2 Feature pruning

The candidate features include many collinear geophysical derivatives. Within each region, features were clustered hierarchically on $1-|\rho_S|$ (Spearman correlation, average linkage) and cut at $|\rho_S| = 0.7$, keeping one representative per cluster chosen by lowest missing rate and then highest variance (Dormann et al., 2013). The rule never sees labels and therefore cannot leak information across folds.

3.3 Models

All models see the same rows and differ only in how they treat the training background or in their loss.

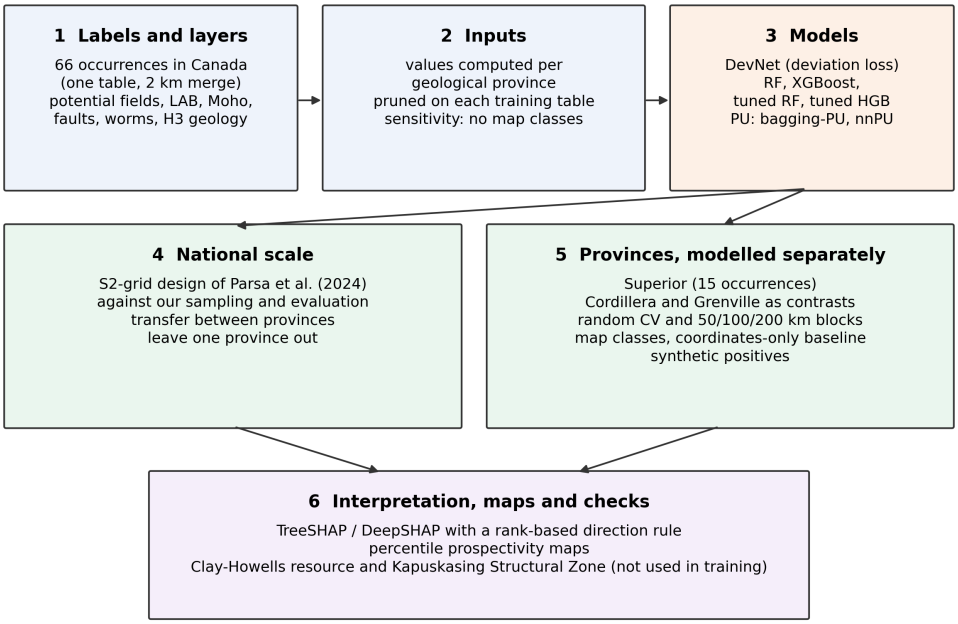


Fig. 3 Workflow. The national experiments motivate modelling the provinces separately. Within a comparison, all models are trained on the same positives and background and scored on the same folds; only the treatment of the background differs.

Random Forest (RF; Breiman, 2001) is the discriminative baseline: positives versus training background labelled 0, 300 trees, $\sqrt{p}$ features per split, balanced class weights, inside a pipeline that fits median imputation and standardisation on the training fold only. With roughly 12 training positives against 5,000 background points, most held-out points receive a predicted probability of exactly zero, so the pooled AUC depends on which trees happen to vote for a given held-out occurrence. We therefore vary the RF seed across folds so that this variance enters the reported intervals.

XGBoost (T. Chen & Guestrin, 2016) adds a gradient-boosted tree ensemble with parameters fixed in advance from a parallel national study of the same commodity rather than tuned here: 200 trees of depth 2, learning rate 0.1, row and column subsampling of 0.8, minimum child weight 3 and L2 and L1 penalties of 2 and 0.5. It is trained like RF, on median-imputed inputs with the positives up-weighted to balance the two classes.

DevNet (Pang et al., 2019) maps each input vector $\mathbf{x}$ to a scalar score $\phi(\mathbf{x})$ through one hidden ReLU layer with dropout. The deviation $d(\mathbf{x}) = (\phi(\mathbf{x}) - \mu_R)/\sigma_R$ is measured against the mean and standard deviation of a fixed reference sample of 50,000 standard-normal draws, and the loss

$$L = (1 - y)\,|d(\mathbf{x})| + y\,\max\{0,\, a - d(\mathbf{x})\} \tag{1}$$

pulls unlabelled background ($y = 0$) towards the reference and pushes positives ($y = 1$) beyond a margin $a = 5$, the value of Pang et al. (2019) (Appendix D, Table D7). Network capacity was set a priori by positive count (64 hidden units, dropout 0.3, weight decay 10^{-4} in Cordillera; 32, 0.4 and 10^{-3} in Superior and Grenville) rather than tuned, because a search over a few tens of positives fits noise. Training uses Adam (Kingma & Ba, 2014) with weight decay, cosine learning-rate annealing over 300 steps and gradient clipping, and each fold score is the rank average of five restarts with different seeds.

Two design choices adapt DevNet to tabular evidence layers with very few positives. First, inputs are mapped to standard-normal marginals by a quantile transformation fitted on each training fold. Many evidence layers are strongly skewed (inverted distances concentrated near 0 or 1, long-tailed textures, 0/1 dummies), and unlike tree ensembles, which are invariant to monotone transformations, multilayer perceptrons are sensitive to such marginal distributions (Gorishniy, Rubachev, Khrulkov, & Babenko, 2021; Grinsztajn, Oyallon, & Varoquaux, 2022). Second, each gradient step uses 16 positives (resampled with replacement) and 240 background rows. The deviation loss is meant to learn what the background looks like; with label-balanced batches of 16 + 16 rows, 300 steps would show each background row about once and each positive about 400 times, whereas background-heavy batches show each background row about 14 times while positive exposure is unchanged. Both choices were settled with diagnostic runs on these provinces using an earlier, coarser encoding of the geology; Appendix D repeats the ablation with the inputs used here.

Bagging-PU (Mordelet & Vert, 2014) trains 200 fully grown decision trees, each on all training positives and an equal-sized bootstrap sample of the unlabelled background, and averages their outputs. It is the standard tree-based PU learner and the closest PU analogue of RF.

nnPU (Kiryo, Niu, du Plessis, & Sugiyama, 2017) minimises the non-negative PU risk with the sigmoid loss, using exactly DevNet's network, quantile-normal inputs, optimiser, number of steps and five-restart ensemble, so that DevNet versus nnPU isolates the PU risk from the deviation loss. nnPU requires the class prior $\pi = P(y = 1)$ in the unlabelled population; we use $\pi = 0.01$ (sensitivity in Appendix A, Table A2). The calibration approach of Elkan and Noto (2008) is not included because under its assumptions the calibrated output is a monotone transform of the positive-versus-background classifier, so its ranking, and every rank metric reported here, equals RF's.

RF-tuned and HGB-tuned give the tree-based baselines a design search of their own. Inside every outer training fold, an inner spatially blocked cross-validation (100 km blocks, three folds, training background purged from inner validation blocks) chooses the RF's minimum leaf size (1 or 10) and number of features per split ($\sqrt{p}$ or $p/3$), or the number of leaves (8 or 31) and minimum leaf size (5 or 40) of a histogram gradient-boosting classifier (Pedregosa et al., 2011) with balanced class weights, 200 iterations, learning rate 0.05 and an L2 penalty of 1. The outer test fold is never used for the choice.

3.4 Evaluation protocol

With three or four positives per validation fold, per-fold AUC is unstable and can be undefined, so out-of-fold scores are pooled (A.P. Bradley, 1997): each held-out point is scored once per repeat, scores are converted to within-fold percentile ranks so that folds are comparable, and a single AUC and average precision (AP) are computed over all folds. Pooling carries a small known negative bias in AUC (Airola et al., 2019; Parker, Günter, & Bedo, 2007). We emphasise AP because, at 0.5% prevalence, it reflects the practical task of concentrating occurrences into a small search area, whereas AUC is dominated by the bulk of easily rejected background (Saito & Rehmsmeier, 2015). Exploration efficiency is also shown as capture-rate curves: the share of held-out occurrences ranked within a given top fraction of the scored pool (Agterberg & Bonham-Carter, 2005; Chung & Fabbri, 2003). Because the evaluation background is a random sample of the domain, this fraction approximates search area. Because unexplored background may host undiscovered occurrences, all metrics are conservative.

Random cross-validation. Repeated stratified 5-fold cross-validation with six repeats gives six pooled AUC and AP values per model, reported as the median and the 2.5–97.5 percentile interval over repeats (labelled 95% interval). All models use identical splits, so they can be compared pairwise.

Spatially blocked cross-validation. Random folds can hold out an occurrence a few kilometres from a training occurrence. Following Roberts et al. (2017) and Valavi, Elith, Lahoz-Monfort, and Guillera-Arroita (2019), labelled points were assigned to square blocks of 50, 100 or 200 km side in Canada Atlas Lambert (EPSG:3978), whole blocks were held out together using stratified group 5-fold splitting (fewer folds if fewer blocks contain positives), and every training-background point falling inside a held-out block was removed from training for that fold. Each of six repeats uses a random offset of the block grid.

Paired comparisons. For each pair of models we average each point's out-of-fold rank over the six repeats and draw 5,000 stratified bootstrap resamples of positives and background (Efron & Tibshirani, 1993), reporting the difference in AP and AUC with its 95% percentile interval and the number of the six repeats in which the first model has the higher AP. With 12–27 positives these comparisons have little power, so we treat a difference as established only when it is consistent across repeats, block sizes and provinces.

Geography-only baseline. To separate geological signal from spatial memorisation, an RF was trained on the projected coordinates of each point only, with no evidence layer, under the same folds and both protocols. Its score measures how much of a protocol's reported skill can be obtained simply by remembering where the training occurrences lie. An L2-regularised logistic regression with balanced class weights serves as a smooth, essentially linear discriminative reference.

3.5 National-scale experiments

Sampling and augmentation. We rebuilt the national S2-grid design of Parsa et al. (2024) with the same geometry library on every level-12 S2 cell (about 4.5 km^2) whose

centre lies in the national domain (1,616,693 cells), with the inputs extracted at the cell centres; the 66 occurrences fall in 66 cells. As in the original study, each occurrence cell was replaced by its four level-13 children, an equal number of negatives was drawn from the 5% of cells with the lowest mean cosine similarity to the positives in min–max scaled feature space, and a random forest of 1,000 trees on min–max scaled inputs was evaluated by test AUC over 50 stratified 70/30 splits of the augmented labels. In our design the same forest, with balanced class weights, is trained on one label per occurrence cell against a random sample of 34,761 cells more than 10 km from any occurrence, and tested on the held-out 30% of occurrence cells against a random 30% of a disjoint sample of 20,855 cells, a realistic prevalence of about 0.3%. To see which element of the original design drives its score, we also changed one element at a time: random instead of cosine-dissimilar negatives, and no augmentation.

Transfer between provinces. The model of province A (its inputs, capacity, occurrences and training background) scored the occurrences and evaluation background of province B, with six seeds; the diagonal, random cross-validation within A, is the provincial result. In leave-one-province-out validation a model trained on the national table without one province group (Superior, Cordillera, Grenville or the rest of Canada) scored that group. To describe the differences between provinces, a random forest (300 trees, 5-fold cross-validation) tried to tell the training background of two province groups apart, and the occurrences of two provinces; and the AUC of each input alone, for a province's occurrences against its evaluation background, gives the direction in which the input marks occurrences there.

3.6 Synthetic positives

To test whether synthetic positives can compensate for label scarcity, each training fold's real positives were augmented with $m \times n_{\text{pos}}$ synthetic positives ($m = 1, 3, 10$) before RF and DevNet were retrained; the held-out positives were always real. Three generators were compared, all fitted in the standardised feature space of the training fold only: Gaussian jitter of real positives ($\sigma = 0.1$ standard deviations); SMOTE-style interpolation between a positive and one of its three nearest positive neighbours (Chawla, Bowyer, Hall, & Kegelmeyer, 2002); and a tabular Wasserstein GAN with gradient penalty (WGAN-GP; Arjovsky, Chintala, & Bottou, 2017; Goodfellow et al., 2014; Gulrajani, Ahmed, Arjovsky, Dumoulin, & Courville, 2017) with an eight-dimensional latent input, two hidden layers of 64 units in generator and critic, five critic updates per generator update, gradient-penalty weight 10 and 2,000 generator updates, trained on the fold's 12 positives. Synthetic geology columns were snapped back to valid one-hot and 0/1 values. To see what the generators produce, each was also fitted to all 15 Superior positives and 150 samples were characterised by the distance to the nearest real positive divided by the median nearest-neighbour distance among real positives (a value near 0 indicates copying), by the share of samples whose nearest labelled neighbour is a positive rather than a background point, and by their DevNet score percentile.

3.7 Interpretation and mapping

Production models were fitted to all positives and the training background of each region. SHAP values (Lundberg & Lee, 2017) were computed with TreeSHAP for RF (Lundberg et al., 2020) and DeepSHAP for DevNet on an explain set of all positives plus evaluation-background points (200 in total), with 100 training-background points as DevNet's reference sample; DevNet's SHAP values refer to its quantile-transformed inputs, which leaves the rank-based directions defined below unaffected. DevNet SHAP values were averaged over the five ensemble members. Importance is mean |SHAP|. The *direction* of a feature's effect is the Spearman correlation between the feature's value and its SHAP value over the explain set (for one-hot dummies, the sign of the mean SHAP difference between presence and absence), with $|\rho| < 0.1$ reported as no clear direction. We do not use the sign of the mean SHAP value, which reflects where the explained points lie relative to the explainer's baseline rather than whether higher feature values raise the score: for RF with balanced class weights the TreeSHAP baseline is near 0.5 while nearly all explained points score near 0, so that rule would label every RF feature as "higher → less prospective". Prospectivity maps classify each model's scores on a 0.1° prediction grid into the top 1%, 5% and 10%, because DevNet's anomaly score and RF's probability are not on a common scale. As an external check, the production models scored Clay-Howells, a drill-tested Fe–REE–Nb prospect with reported resources on the KSZ (Ontario Geological Survey, 2011) that is not among the training labels, at its recorded location.

4 Results

4.1 National scale

4.1.1 Sampling and augmentation in the S2-grid design

Rebuilt on our national data, the design of Parsa et al. (2024) gives a test AUC of 1.000 (Table 2), as high as the published 0.91–0.96 or higher. With one label per occurrence cell, random background and a realistic prevalence, the same forest reaches an AUC of 0.903 and an AP of 0.292. Each of the two elements of the original design suffices on its own for a near-perfect score. Splitting augmented children at random leaves a sibling of 97% of the test positives in the training set, and with random negatives the AUC stays at 0.996; cosine-dissimilar negatives without any augmentation give 0.997, because the ground least similar to the occurrences is easy to reject. Such negatives also make a worse model: trained against them rather than against random negatives in an otherwise identical design, the forest's AP at realistic prevalence falls from 0.184 to 0.091.

4.1.2 Transferability between provinces

A province's model does not predict another province (Fig. 4). Within each province, random cross-validation gives an AP of 0.13–0.45 (nnPU excluded); applied to another province, the tree ensembles reach 0.004–0.012, about the random-ranking values of 0.004–0.009, and in 20 of the 30 model–province pairs the AUC is below 0.5: the

Table 2 The national S2-grid design of Parsa et al. (2024) and ours, with the same random forest (1,000 trees, min–max inputs): mean test AUC and AP over 50 stratified 70/30 splits. Test set: occurrence share among the test points.

Design	Positive labels	Negatives	Split	Test set	AUC	AP
Parsa et al. (2024)	four L13 children per occurrence cell (264)	264 cells from the 5% least cosine-similar	child labels, random 70/30	balanced (50%)	1.000	1.000
This study	one per occurrence cell (66)	34,761 random cells >10 km from occurrences	occurrence cells, random 70/30	realistic (0.3%)	0.903	0.292

model ranks the other province's occurrences below its background. The exception is DevNet trained in Cordillera and applied to Superior (AUC 0.79, AP 0.038 against 0.005). Leaving a province out of a national model gives the same picture: held-out Grenville is ranked at or below chance by every model (AUC 0.32–0.68, AP $\leq$ 0.026, against 0.004 for a random ranking), and held-out Superior reaches an AP of at most 0.051 (random ranking 0.005).

The provinces are distinct domains in the inputs. A random forest separates the training background of any two province groups with an AUC of 0.999–1.000, and the occurrences of Superior, Cordillera and Grenville from one another with 0.99–1.00. Several inputs mark occurrences in opposite directions in different provinces. Taken alone, LAB depth separates Superior occurrences from their background with an AUC of 0.22 (shallower LAB at occurrences) but Cordilleran ones with 0.67 (deeper), Moho depth with 0.35 and 0.78, and distance to regional greenstone with 0.23 and 0.63; deep-source magnetics marks Cordilleran and Grenville occurrences in opposite directions (0.25 and 0.81). The rest of the study therefore models each province separately.

4.2 Provincial models

4.2.1 Model comparison under random cross-validation

In Superior all models except nnPU rank held-out occurrences far above chance (Table 3): AP is 0.22–0.45 against a random-ranking value of 0.005, and AUC is 0.97–0.98. DevNet has the highest median AP (0.445) and RF the highest AUC (0.983). DevNet's AP exceeds RF's, XGBoost's (0.275), bagging-PU's and nnPU's in all six repeated splits, by 0.18–0.26 on repeat-averaged ranks. nnPU, which shares DevNet's network and inputs, reaches an AP of only 0.152 and an AUC of 0.838.

The contrasting regions behave differently. In Cordillera, bagging-PU and RF lead (AP 0.433 and 0.427), and DevNet (0.341; lower than RF in all six repeats) and XGBoost (0.347) trail. In Grenville all models are weak; XGBoost leads (0.305), followed by RF (0.257), both ahead of DevNet (0.125) in all six repeats. nnPU fails in both provinces (AUC 0.54–0.55). That Superior is the best-predicted of the three provinces agrees with the province-level success rates of Parsa et al. (2024).

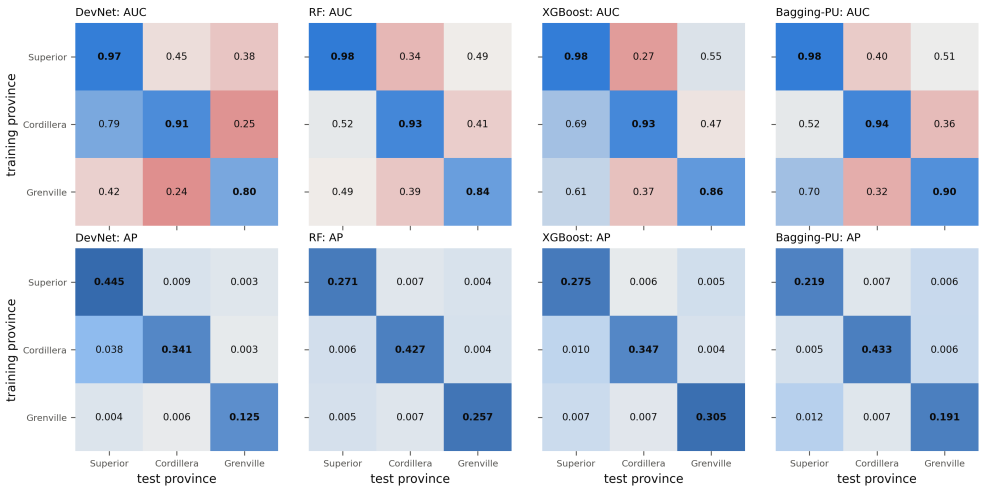


Fig. 4 Transfer between provinces. Top: AUC; bottom: AP, coloured by AP divided by the test province's prevalence (log scale). Rows: province whose model (inputs, capacity, occurrences, training background) is applied; columns: province whose occurrences and evaluation background are scored (median of six seeds). The diagonal is random cross-validation within the province (Table 3).

Table 3 Pooled out-of-fold AUC and AP (median and 95% interval over six repeated 5-fold random splits). Bold: best of the five models in each region.

Region (n_{pos})	Model	AUC	AP
Superior (15)	DevNet	0.972	**0.445** [0.404, 0.474]
	RF	**0.983**	0.271 [0.165, 0.289]
	XGBoost	0.976	0.275 [0.257, 0.353]
	Bagging-PU	0.976	0.219 [0.158, 0.264]
	nnPU	0.838	0.152 [0.125, 0.191]
Cordillera (27)	DevNet	0.914	0.341 [0.299, 0.394]
	RF	0.931	0.427 [0.368, 0.516]
	XGBoost	0.927	0.347 [0.316, 0.380]
	Bagging-PU	**0.936**	**0.433** [0.304, 0.507]
	nnPU	0.553	0.022 [0.019, 0.029]
Grenville (12)	DevNet	0.799	0.125 [0.054, 0.171]
	RF	0.838	0.257 [0.182, 0.489]
	XGBoost	0.857	**0.305** [0.207, 0.494]
	Bagging-PU	**0.898**	0.191 [0.172, 0.255]
	nnPU	0.542	0.008 [0.006, 0.009]

4.2.2 Spatially blocked cross-validation

Holding out whole blocks widens the gap between DevNet and the tree-based and PU learners in Superior (Fig. 5; Table 4). DevNet's AP is 0.418, 0.343 and 0.252 with 50, 100 and 200 km blocks. RF's falls from 0.271 under random splits to 0.192, 0.168 and 0.151, XGBoost's from 0.275 to 0.202, 0.123 and 0.131, and bagging-PU's to 0.148, 0.129 and 0.090. DevNet has the higher AP than RF, bagging-PU and nnPU in all

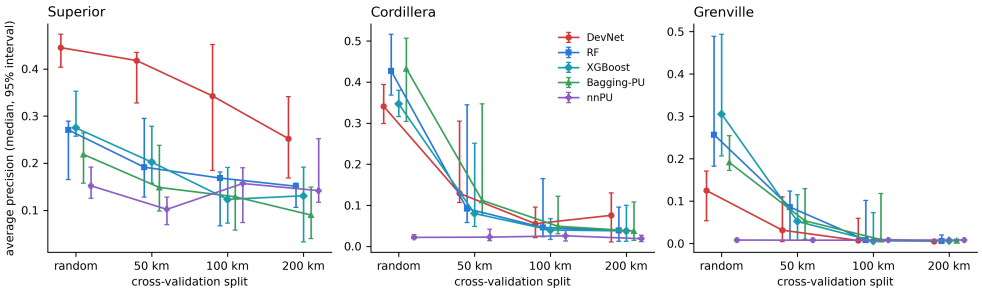


Fig. 5 Average precision under random and spatially blocked cross-validation (median and 95% interval over six repeats). The random-ranking AP is 0.005 (Superior), 0.009 (Cordillera) and 0.004 (Grenville).

six repeats at every block size, and than XGBoost in 17 of the 18, by +0.18 to +0.24 against RF, +0.23 to +0.31 against XGBoost, +0.25 to +0.33 against bagging-PU and +0.19 to +0.41 against nnPU. With 15 occurrences, however, the 95% intervals of the differences from RF reach down to about zero, so the margin over RF is consistent but not statistically conclusive; the margins over bagging-PU and nnPU are clearer. AUC tells a partly different story with 200 km blocks, where RF (0.880), bagging-PU (0.855) and XGBoost (0.826) exceed DevNet (0.783; Appendix A, Table A1): DevNet ranks most held-out occurrences very high but misses a few almost entirely, notably Kapuskasing and Nemegosenda Lake.

Capture-rate curves (Fig. 6) show where the AP differences arise. Under random splits DevNet and RF capture about half of the held-out occurrences in the top 1% of the ranked pool (53% and 50%), and RF and bagging-PU capture all of them by the top 10% (DevNet 93%). With 50 and 100 km blocks DevNet keeps its top-1% capture (53% and 47%) and captures 90–93% by the top 10%, whereas RF captures 43% in the top 1% and 83–90% by the top 10%, and bagging-PU 30% and 83–90%. With 200 km blocks every model captures 47% or less in the top 1%, and by the top 10% RF and bagging-PU (70%) slightly exceed DevNet (63%).

In Cordillera every model degrades sharply under blocking (AP $\leq$ 0.17 at 50 km and $\leq$ 0.09 at 100–200 km), and in Grenville AP approaches the random-ranking value ($\leq$ 0.009 at 100–200 km). In these regions no difference between models is of practical size: their random-split skill owes much to spatial proximity between training and test occurrences (Section 4.2.5).

4.2.3 Contribution of geological map classes

A single map class carries much of the skill in Superior. Removing the alkalic-intrusion column lowers DevNet's AP to 0.345, 0.256, 0.219 and 0.154 under random splits and 50, 100 and 200 km blocks, and RF's to 0.230, 0.126, 0.096 and 0.084: this one class, which covers 6 of the 15 Superior occurrences and 0.5% of the background, accounts for over a third of DevNet's spatially blocked AP. DevNet without it still has a higher AP than RF with it at every block size. Withholding all point-level map classes costs

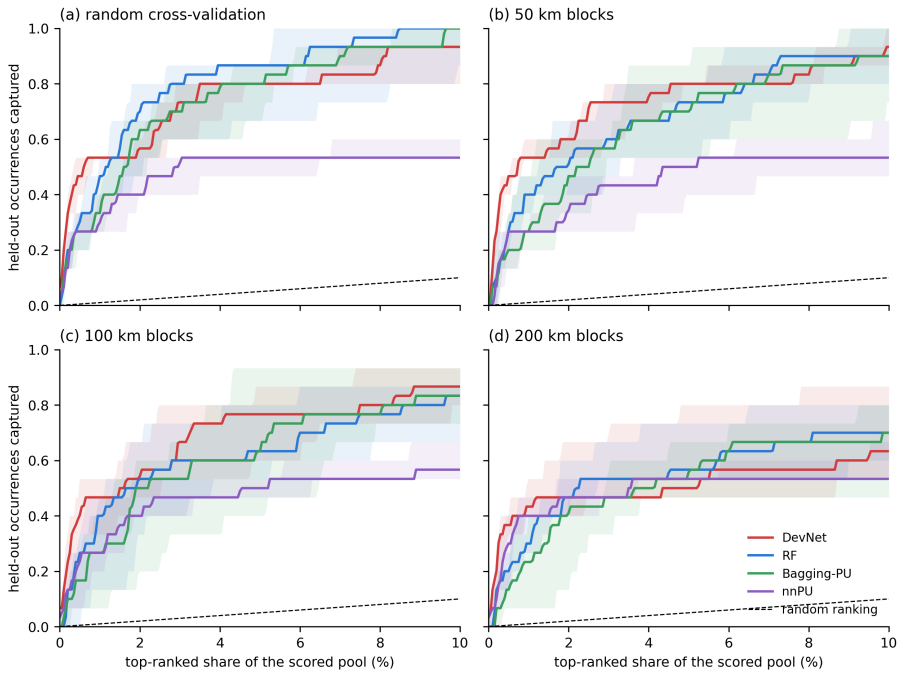


Fig. 6 Superior capture-rate curves: share of held-out occurrences ranked within the top fraction of the scored pool (a proxy for search area) under (a) random cross-validation and (b) 50 km, (c) 100 km and (d) 200 km spatial blocks. Lines: median over six repeats; bands: range. Dashed: random ranking.

DevNet and RF about half or more of their AP: DevNet's falls to 0.183 under random splits and 0.133, 0.109 and 0.079 under blocks, RF's to 0.138 and 0.084, 0.069 and 0.051. Without map classes DevNet still has the highest AP under every block size, narrowly ahead of bagging-PU (0.130, 0.079 and 0.065), while bagging-PU leads under random splits (0.225) and nnPU fails (AP 0.007).

4.2.4 Tuned baselines

Tuning did not help the tree-based learners in Superior (Table 5). RF-tuned reached an AP of 0.263, 0.163, 0.142 and 0.117 under random splits and 50, 100 and 200 km blocks, and HGB-tuned 0.200, 0.118, 0.087 and 0.063, at or below the default RF; DevNet had the higher AP than both in all six repeats under every protocol. XGBoost, with parameters fixed in advance, behaves like these baselines (0.275, 0.202, 0.123 and 0.131; Table 4). In Cordillera, HGB-tuned has the highest random-split AP (0.470) and in Grenville the default RF (0.257); under blocks all models are close to chance in both provinces.

Table 4 Average precision (median over six repeats) under random and spatially blocked cross-validation. Bold: highest AP in each row. AUC for the same runs is given in Appendix A (Table A1).

Region	Split	DevNet	RF	XGBoost	Bagging-PU	nnPU
Superior	random	**0.445**	0.271	0.275	0.219	0.152
	50 km	**0.418**	0.192	0.202	0.148	0.102
	100 km	**0.343**	0.168	0.123	0.129	0.157
	200 km	**0.252**	0.151	0.131	0.090	0.142
Cordillera	random	0.341	0.427	0.347	**0.433**	0.022
	50 km	**0.129**	0.092	0.080	0.112	0.023
	100 km	**0.055**	0.046	0.040	0.049	0.026
	200 km	**0.076**	0.039	0.038	0.038	0.019
Grenville	random	0.125	0.257	**0.305**	0.191	0.008
	50 km	0.031	**0.086**	0.052	0.053	0.008
	100 km	0.007	0.008	0.006	**0.009**	0.008
	200 km	0.005	0.006	0.006	0.007	**0.008**

Table 5 Tuned tree-based baselines: AP (median over six repeats). RF and HGB (tuned): hyperparameters chosen inside each outer training fold by inner 100 km spatial cross-validation. Bold: highest AP in each row.

Region	Split	DevNet	RF	RF (tuned)	HGB (tuned)
Superior	random	**0.445**	0.271	0.263	0.200
	50 km	**0.418**	0.192	0.163	0.118
	100 km	**0.343**	0.168	0.142	0.087
	200 km	**0.252**	0.151	0.117	0.063
Cordillera	random	0.341	0.427	0.435	**0.470**
	50 km	**0.129**	0.092	0.110	0.101
	100 km	**0.055**	0.046	0.048	0.036
	200 km	**0.076**	0.039	0.036	0.037
Grenville	random	0.125	**0.257**	0.164	0.203
	50 km	0.031	0.086	0.067	**0.097**
	100 km	0.007	0.008	**0.009**	0.006
	200 km	0.005	**0.006**	0.005	0.006

4.2.5 Spatial proximity and the geography-only baseline

A random forest given only coordinates reaches an AP of 0.507 in Cordillera under random cross-validation, higher than any evidence-based model (at most 0.470, HGB-tuned), and 0.249 in Grenville, close to RF (0.257) and below only XGBoost (0.305; Table 6). In these two provinces the random-split skill of the tree-based models can therefore be reproduced without any geology: training occurrences cluster, a random split leaves most held-out occurrences close to a training occurrence, and a flexible learner can exploit that proximity, through the coordinates themselves or through spatially smooth layers that act as surrogate coordinates. Under 100 km blocks the coordinate-only model collapses (0.052 and 0.075), as do the evidence-based models (≤ 0.062 and ≤ 0.009).

Superior is the exception. Coordinates alone reach only 0.157 under random splits and 0.018 under blocking, far below every evidence-based model except nnPU, so

Table 6 Geography-only and linear baselines: AP and AUC (median over six repeats) under random cross-validation and 100 km spatial blocks, on identical folds. RF, coordinates only: random forest trained on projected coordinates without any evidence layer.

		Random CV		100 km blocks	
Region	Model	AP	AUC	AP	AUC
Superior	RF, coordinates only	0.157	0.782	0.018	0.750
	RF, all features	0.271	0.983	0.168	0.904
	Logistic regression	0.374	0.924	0.219	0.942
	DevNet	0.445	0.972	0.343	0.950
Cordillera	RF, coordinates only	0.507	0.922	0.052	0.727
	RF, all features	0.427	0.931	0.046	0.776
	Logistic regression	0.210	0.916	0.062	0.826
	DevNet	0.341	0.914	0.055	0.790
Grenville	RF, coordinates only	0.249	0.821	0.075	0.750
	RF, all features	0.257	0.838	0.008	0.538
	Logistic regression	0.037	0.766	0.005	0.624
	DevNet	0.125	0.799	0.007	0.661

the Superior scores reflect a signature in the evidence rather than location. DevNet uses the evidence better than an L2-regularised logistic regression in every province (random splits: 0.445, 0.341 and 0.125 against 0.374, 0.210 and 0.037; 100 km blocks in Superior: 0.343 against 0.219).

4.2.6 Synthetic positives

The generators behave very differently (Table 8; Appendix B, Fig. B1). Jitter and SMOTE samples lie close to the real positives (median nearest-neighbour ratio 0.10 and 0.24) and almost all have a real positive as their nearest labelled neighbour (100% and 97%): they are near-copies of the training occurrences. WGAN-GP samples are as far from the real positives as the real positives are from one another (ratio 0.91), but for 35% of them the nearest labelled point is a background point: with 15 training points the generator spreads probability mass into background-like parts of feature space. All three sets of samples receive high DevNet scores (median percentile 0.98).

The folds are identical with and without augmentation (Table 7; Fig. 7). For DevNet no generator helps: with augmentation its AP lies between 0.374 and 0.444, against 0.445 without. For RF, jitter raises the AP from 0.271 to 0.301–0.327, SMOTE lowers it (0.194–0.232) and WGAN-GP leaves it at or below the unaugmented value (0.235–0.267). The best augmented RF (jitter 10×, AP 0.327) remains well below DevNet without augmentation (0.445). Under spatial blocking (Appendix B, Table B3) WGAN-GP at 10× raises RF's AP with 100 km blocks from 0.168 to 0.219, still below unaugmented DevNet (0.343), but with 200 km blocks every generator lowers RF's AP (0.100–0.137 against 0.151). DevNet's AP under blocking changes little (0.321–0.346 with 100 km and 0.246–0.260 with 200 km blocks).

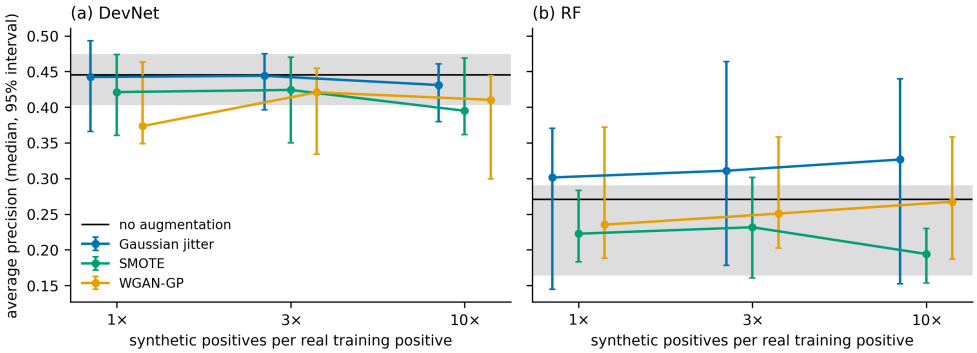


Fig. 7 Superior average precision with fold-safe synthetic positives, by generator and number of synthetic positives per real training positive. Line and grey band: no augmentation (median and 95% interval).

Table 7 Fold-safe positive augmentation, Superior (random cross-validation, identical folds; held-out positives always real): AUC and AP (median and 95% interval over six repeats).

Model	Generator	m	AUC	AP
DevNet	none	–	0.972	0.445 [0.404, 0.474]
	Gaussian jitter	1×	0.964	0.442 [0.366, 0.493]
	Gaussian jitter	3×	0.973	0.444 [0.396, 0.475]
	Gaussian jitter	10×	0.974	0.431 [0.380, 0.461]
	SMOTE	1×	0.974	0.421 [0.360, 0.474]
	SMOTE	3×	0.972	0.424 [0.350, 0.470]
	SMOTE	10×	0.975	0.395 [0.362, 0.469]
	WGAN-GP	1×	0.968	0.374 [0.349, 0.464]
	WGAN-GP	3×	0.970	0.421 [0.334, 0.454]
	WGAN-GP	10×	0.964	0.410 [0.299, 0.444]
RF	none	–	0.983	0.271 [0.165, 0.289]
	Gaussian jitter	1×	0.973	0.301 [0.144, 0.371]
	Gaussian jitter	3×	0.983	0.311 [0.178, 0.464]
	Gaussian jitter	10×	0.975	0.327 [0.152, 0.440]
	SMOTE	1×	0.980	0.223 [0.183, 0.283]
	SMOTE	3×	0.984	0.232 [0.161, 0.301]
	SMOTE	10×	0.981	0.194 [0.154, 0.230]
	WGAN-GP	1×	0.963	0.235 [0.188, 0.372]
	WGAN-GP	3×	0.979	0.251 [0.202, 0.358]
	WGAN-GP	10×	0.985	0.267 [0.187, 0.358]

4.2.7 Model interpretation

With the rank-based direction rule, RF and DevNet agree on the sign of the effect for all ten of their jointly most important features in each province (Table 9; Appendix C). Read from the sign of the mean SHAP value instead, the two models would appear to disagree on many of these features, an artefact of the two explainers' different baselines (Section 3.7).

Table 8 What the generators produce (fitted to all 15 Superior positives; 150 samples each). NN ratio: distance to the nearest real positive divided by the median nearest-neighbour distance among real positives (0 = copy). Nearest labelled neighbour: share of samples whose nearest point among positives and training background is a positive. DevNet percentile: median score percentile among the training background under the production DevNet.

Source	NN ratio, median	NN ratio, P10	NN is positive	DevNet pct.
Real occurrences (leave-one-out)	1.00	0.55	–	0.995
Gaussian jitter	0.10	0.08	100%	0.984
SMOTE	0.24	0.06	97%	0.982
WGAN-GP	0.91	0.58	65%	0.985
Background (300 random points)	1.51	–	–	0.500

Table 9 Superior: the ten jointly most important features (mean |SHAP|, normalised per model). Rank among 48 features; direction ↑/↓: higher values (or presence) raise/lower the score, with Spearman ρ between feature value and SHAP value.

Feature	DevNet rank	DevNet dir.	RF rank	RF dir.
Deep-source RTP HGM	9	↓ (-0.95)	1	↓ (-0.74)
Lith.: Igneous Extrusive	1	↑ (+0.53)	34	↑ (+0.53)
LAB depth	8	↓ (-0.94)	2	↓ (-0.90)
Distance to regional greenstone	6	↓ (-0.89)	3	↓ (-0.90)
Age contact (H3 cell)	2	↑ (+0.50)	31	↑ (+0.50)
TMI	11	↑ (+0.92)	5	↑ (+0.86)
Lith.: Igneous Intrusive Alkalic	3	↑ (+0.34)	20	↑ (+0.34)
Fault proximity	15	↑ (+0.90)	4	↑ (+0.84)
Bouguer anomaly (BA) GLCM correlation	7	↓ (-0.88)	16	↓ (-0.90)
Deep magnetic-worm proximity	10	↓ (-0.92)	11	↓ (-0.93)

The two models agree on directions but weight the evidence differently (Fig. 8). In Superior, DevNet ranks five geological classes of the cell first: extrusive (greenstone) lithology, an age contact within the cell and alkalic intrusive lithology raise the score, and Archean era and tonalite lower it; it then ranks proximity to regional greenstone, Bouguer GLCM correlation (lower values more prospective) and LAB depth. RF ranks the horizontal-gradient magnitude of deep-source magnetics first (lower values more prospective), then LAB depth, proximity to regional greenstone and fault proximity, and gives the geological classes low ranks (20th–34th of 48). Both models score ground as more prospective where the LAB is shallower, close to greenstone belts and faults, where TMI is high, and farther from deep magnetic worms. In Cordillera both models treat Mesozoic terranes and ground far from high-grade metamorphic units as unfavourable and Proterozoic terranes as favourable. In Grenville both rank deep-source magnetics and distance to regional age contacts among their four most important features, but given the weak predictive skill there these attributions describe the signal the models attempted to use rather than a reliable control.

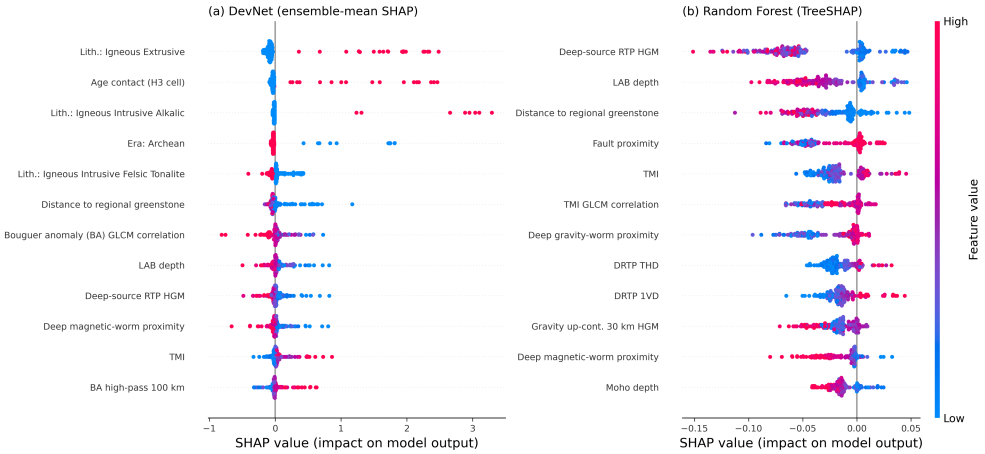


Fig. 8 SHAP summaries for Superior: (a) DevNet (DeepSHAP on its quantile-transformed inputs, mean over the five ensemble members) and (b) RF (TreeSHAP). Each point is one explained sample; colour is the feature value. SHAP magnitudes are not comparable between models (unbounded anomaly score vs probability).

4.2.8 Prospectivity maps and external validation

The production DevNet and RF maps (Fig. 9) both concentrate their highest classes in the southern Superior Province, around the eastern-Abitibi cluster and across the Abitibi and Wawa subprovinces, but their rankings of the province are only moderately similar (Spearman correlation over the grid 0.37). RF assigns a probability of exactly zero to 83% of the grid, so its classes are drawn from the remaining nodes. In the Kapuskasing area (Fig. 10) DevNet places its top 1% mainly in the Abitibi Subprovince east of the KSZ; along the KSZ itself, the top 5% and 10% classes form discontinuous north-north-east trending patches on its western side. Clay-Howells, the untrained Fe–REE–Nb resource on the KSZ, lies 79 km from the nearest training occurrence; among the province's 8,000 random background points the production models place it at the 96.9th (DevNet) and 99.4th (RF) percentile.

5 Discussion

5.1 National-scale models and transferability

The national experiments explain why we model the provinces separately. First, the high test scores of the S2-grid design do not measure prediction on unexplored ground. Random splitting of augmented child cells leaves near-copies of the test positives in the training set, and negatives drawn from the least similar ground remove from evaluation exactly the ground on which a prospectivity model is hard to get right; either element alone gives a near-perfect AUC, and with one label per occurrence and random background at a realistic prevalence the same forest reaches an AP of 0.29. Second, a larger, pooled label set does not buy transferability. The provinces

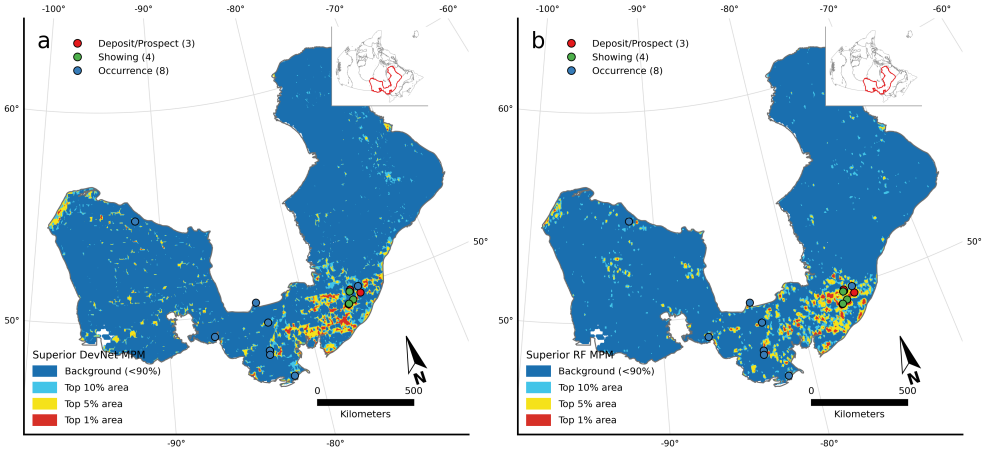


Fig. 9 Superior prospectivity maps from the production (a) DevNet and (b) RF models, classified by score percentile over the mapped area (top 1%, 5% and 10%), with known occurrences by development class (Canada Atlas Lambert). Scores on the 0.1° prediction grid were interpolated to a 2 km raster.

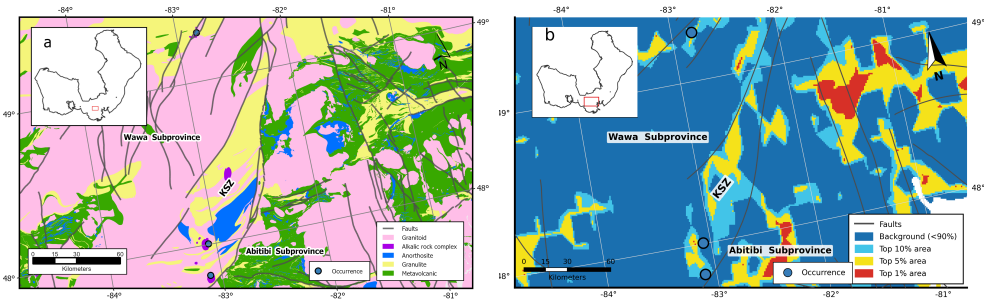


Fig. 10 Kapuskasing area. (a) Regional geology with the Kapuskasing Structural Zone (KSZ) and alkalic rock complexes (Gittins et al., 1967; Percival & West, 1994), compiled from Heather, Percival, Moser, and Bleeker (1995) and Montsion, Thurston, and Ayer (2018). (b) DevNet top 1%, 5% and 10% classes (province-wide percentiles), with faults from the compilation used as an evidence layer, on the same frame as (a).

are separable domains, several layers mark occurrences in opposite directions (LAB and Moho depth, distance to greenstone), and models trained in one province rank another province's occurrences at or below chance. In the terms of Meyer and Pebesma (2021), a province lies outside the area of applicability of a model trained on another, and pooling widens the training domain without making the model applicable to each part of it. A national model has to learn several expressions of the mineral system at once from 66 occurrences; a provincial model has to learn one.

5.2 Performance of the deviation network

In Superior DevNet ranks held-out occurrences better than RF, XGBoost, tuned RF, tuned gradient boosting, bagging-PU and nnPU in every repeat of every protocol, with one exception against XGBoost, and its lead is largest when the test requires extrapolation: RF, XGBoost and bagging-PU, which learn sharp decision boundaries around the training positives, lose 26–59% of their AP once the nearest training occurrence is 50–200 km away, whereas DevNet keeps most of its AP to 100 km. With 15 positives the margin over RF is not statistically conclusive on its own, and DevNet's design was chosen with these provinces in view; but the lead is consistent across all 18 repeat–block combinations and it survives a design search for the tree-based learners.

The design of the network matters (Appendix D, Table D6). Its score is a sum of a few ReLU ridges in the space of its inputs, so its resolution along each feature is set by how that feature is scaled; on z-scored, strongly skewed layers a handful of extreme values occupy most of the input range and the bulk of the ground is compressed into a narrow interval. Quantile-normal inputs give every feature a comparable, rank-based scale, the information that trees exploit through their thresholds, while keeping the score smooth, and background-heavy batches let the loss see much more of the background per update. Together they raise DevNet's random-split AP in Superior from 0.358 to 0.445. A smooth score of this kind is rewarded when held-out occurrences lie far from the training ones, whereas the thresholds of tree ensembles carve out the neighbourhood of each training positive.

Among the learners that treat the background as unlabelled, the deviation loss is the one that works. nnPU, which shares DevNet's network and inputs and differs only in its risk, trails it by 0.19–0.41 in AP under blocking, fails in Cordillera and Grenville and without map classes, and depends on a class prior that is unknown in practice (AP 0.13–0.26 for π between 0.005 and 0.05); bagging-PU trails it under every protocol.

The Cordilleran and Grenville results bound the conclusion. Where positives are comparatively plentiful and clustered (Cordillera), or the geophysical signal is weak (Grenville), the tree-based learners are as good or better under random splits, a coordinate-only forest matches or nearly matches them, and no model transfers between blocks. DevNet is therefore not a universal replacement for discriminative baselines; its advantage is specific to the combination of very few positives, a shared geological setting and the need to extrapolate within it.

5.3 Geological map classes: evidence and lookup

The map classes move every model more than the choice of model does. Withholding them costs DevNet and RF about half or more of their AP, and a single class, alkalic intrusive rock, carries over a third of DevNet's blocked AP. Six of the 15 Superior occurrences lie on cells mapped as alkalic intrusive rock, which covers 0.5% of the background; for these occurrences the map already marks the target. The age-contact flag of the containing cell partly does the same, because a Proterozoic complex mapped within Archean crust creates an age contact, and DevNet ranks it second.

Whether such lookup is acceptable depends on the use. For targeting within mapped complexes, or for ranking known complexes by their geophysical setting, a

mapped alkaline intrusion is legitimate and strong evidence, and the full inputs are appropriate. For finding complexes that the 1:5,000,000 map does not show, it is not, and the relevant skill is that without map classes: an AP of 0.08–0.13 under spatial blocking, 16–27 times the random-ranking value but well below what the full inputs suggest. Comparisons between studies, and claims about how well a model "discovers" occurrences, should state which of the two is being measured. In both settings DevNet is at or near the top under blocking; its absolute skill, however, depends strongly on the map classes.

5.4 Differences between provinces

Grenville is the weakest of the three provinces for every model and Superior the strongest. The evidence layers are the same national compilations everywhere and almost complete (about 0.01% of values missing in each province), so the contrast does not reflect data coverage. It follows instead from the label sets and from the geology they sample.

First, the occurrences are less independent than their count suggests, most of all in Cordillera and Grenville: there, more than half of the occurrences lie within 10 km of another (median distance to the nearest other occurrence 7 km, against 31 km in Superior). Under a random split such neighbours fall on both sides, which is why coordinates alone score well in these two provinces (Section 4.2.5) and why their skill collapses under blocking.

Second, the Grenville label set mixes carbonatites of different ages and tectonic settings. Crevier is a late-Grenvillian alkaline–carbonatite intrusion of about 957 Ma (Groulier et al., 2020). St-Honoré, with Terres Rompues on the same complex, was emplaced at about 580 Ma in the Saguenay graben of the St. Lawrence rift system (Néron, Bédard, & Gaboury, 2018). The Lavergne–Springer occurrences lie near Lake Nipissing at the western end of the Ottawa–Bonnechère graben, and the Cantley, Quinnville and Templeton showings near Gatineau belong to the calcite-rich veins and dykes of the Grenville marble belts, whose magmatic origin has long been debated (Moecher, Anderson, Cook, & Mezger, 1997). Age diversity alone is not the problem, since the Superior occurrences also span several magmatic episodes (Rukhlov & Bell, 2010). They were, however, all emplaced into the same stable cratonic lithosphere, whose margins and crustal-scale structures the present-day LAB, Moho and potential-field layers still record. The Grenville crust was reworked by the Grenvillian orogeny and by later rifting, so its present-day geophysics need not record a common setting for occurrences that formed in different ones. The mapped geology adds noise of its own: the twelve Grenville occurrences fall on Archean, Proterozoic and Paleozoic map units, because the bedrock map records reworked protoliths and the Paleozoic cover over St-Honoré rather than the carbonatites, and none of them lies on a cell mapped as alkalic intrusive rock, whereas about 5% of the Grenville background does. The same lack of a common setting, at a larger scale, is why models do not transfer between provinces (Section 4.1.2).

Missing or coarse geophysics therefore explains little of the contrast; the number, independence and geological coherence of the labels explain most of it. A model for

Grenville would need either more occurrences of one carbonatite type or separate models for rift-related and orogenic types, which the present label set cannot support.

5.5 Synthetic positives and label scarcity

Synthetic positives are an attractive response to label scarcity, and GAN-generated labels have been recommended for exactly this commodity (Parsa et al., 2025). Our results counsel caution. A generator fitted to a dozen training positives has two options. It can stay close to them, in which case it adds near-duplicates that carry no new information (jitter and SMOTE, nearest-neighbour ratios of 0.1–0.2); or it can spread out, in which case it must guess where unseen occurrences lie and, having no information about that, places part of its mass in background-like feature space (WGAN-GP, 35% of samples nearest to a background point). Neither option supplies the information that is actually missing, namely what undiscovered occurrences look like. The gains for RF from jitter are better explained as smoothing: synthetic points around the training positives make more trees vote for nearby held-out points, which relieves RF's degenerate surface of exactly-zero probabilities rather than teaching it new geology. Consistent with this reading, they vanish and turn into losses with 200 km blocks, where held-out occurrences are far from every training one, and DevNet, whose loss already produces smooth scores, gains nothing from the same points. Where synthetic positives are used, they must be generated inside each training fold from that fold's positives only, and performance must be measured on real held-out occurrences; otherwise a synthetic copy of a training occurrence in the test set reproduces the sibling problem of Section 4.1.1 in another form.

5.6 Validation design and reported performance

Four of our findings concern evaluation rather than models. First, with tens of positives a single cross-validation split gives a noisy estimate: re-splitting the same Superior data moves each model's AP over a range of 0.07–0.17 (Table 3), as large as many of the differences between models. Second, random splits overstate skill wherever occurrences cluster, and the size of the overstatement differs between models, so random splits can also reverse a model ranking (Section 4.2.2). Third, rebuilt nationally, the S2-grid design gives a test AUC of 1.000, and two of its elements each suffice for it: splitting augmented child cells at random (0.996 with random negatives), because almost every test positive has a sibling, sharing the coarse deep-architecture values and the map classes exactly and the fine-scale values nearly, on the training side; and drawing negatives from the least similar ground (0.997 without augmentation). In this design the held-out test asks the model to recognise a location it has already seen, and the comparison of training and test AUC, which Parsa et al. (2024) used to argue that their models were not over-fitted, becomes uninformative because the test set is not independent of the training set. Selecting negatives from the least similar 5% of the ground adds a second, independent source of optimism: it removes from evaluation exactly the ground on which a prospectivity model is hardest to get right. Fourth, the geological map classes can supply a large share of the reported skill by marking known complexes (Section 5.3), which no split design removes.

We stress what the national experiment does and does not show. It uses our inputs, labels and models, not those of Parsa et al. (2024), and it does not establish what their numbers would have been under a grouped split. It does show that, on national Canadian carbonatite data, the published design produces a test AUC of 1.000, whereas the same forest with one label per occurrence, random background and a realistic prevalence reaches an AUC of 0.903 and an AP of 0.292, and that a model trained in one province does not carry over to another. Neither the high AUC nor its small spread over splits can be taken as evidence of generalisation to unexplored ground. Parsa et al. (2025) describe the same stratified random 70/30 split for a workflow that adds GAN-generated labels; their description does not state whether synthetic labels derived from an original label were kept on the same side of the split as that label. If they were not, the caveat applies with greater force. For prospectivity studies under extreme label scarcity we therefore recommend (i) grouping every augmented or near-duplicate label with its source occurrence before any split, (ii) drawing evaluation negatives at random from the domain rather than from the ground least similar to the positives, (iii) reporting AP with intervals over repeated splits rather than a single AUC, (iv) adding a spatially blocked validation in which training background is purged from held-out blocks, (v) reporting a geography-only baseline, so that readers can see how much of a score is location, (vi) reporting results with and without point-level map classes, so that readers can see how much of a score is map lookup, and (vii) for models pooled across geological provinces, testing transfer to held-out provinces and computing extent-dependent inputs within fixed geological units.

5.7 Geological interpretation

The drivers on which RF and DevNet agree in Superior describe a coherent setting. Shallower LAB, together with the strong ranking of deep-source magnetic gradients, points to the thinner lithosphere at the edges of the thick Superior root, where small-degree melting of metasomatised mantle and access along lithosphere-scale structures are most likely (Griffin et al., 2013; Groves & Santosh, 2021; Hoggard et al., 2020); a mantle conductor beneath the southern Abitibi is consistent with such metasomatism (Snyder et al., 2021). Fault proximity and high magnetic intensity capture the structures that focus ascent and the magnetite-bearing intrusions themselves (Thomas et al., 2016). The negative effect of proximity to deep magnetic worms is less intuitive; it may reflect the concentration of deep magnetic contacts along terrane boundaries that are not themselves carbonatite hosts, and should not be over-interpreted.

The strong positive effect of greenstone lithology and of proximity to regional greenstone units needs more care. Eight of the 15 occurrences belong to the eastern-Abitibi cluster in a greenstone-dominated subprovince, so the association may partly record where the largest cluster lies rather than a control shared by the dispersed occurrences; the dispersed occurrences that DevNet ranks lowest under blocking, Kapuskasing and Nemegosenda Lake, lie in the high-grade and alkaline terrain of the KSZ rather than in a greenstone belt. Consistent with this, DevNet's map follows the KSZ only in discontinuous patches, while the untrained Clay-Howells resource on the KSZ still ranks at its 97th percentile. A model whose top classes are tied to one

greenstone-hosted cluster is useful for ranking ground of that type, but it should not be expected to recognise alkaline complexes in other settings as well.

5.8 Limitations

Fifteen positives allow only weak statistical statements: our firm results are those that are consistent across repeats, block sizes, comparisons and provinces. DevNet's input representation and batch composition were chosen with diagnostic experiments on these three provinces, so part of the final configuration's advantage may reflect that choice; confirmation on an independent province is needed, and the Churchill Province, the obvious candidate, holds too few carbonatite-hosted occurrences after deduplication for a meaningful test. The geological encoding used here replaced a coarser grouping of lithologies after that grouping had been evaluated; the change raised the AP of every model except nnPU under random splits and 50–100 km blocks but not with 200 km blocks. The evaluation background may contain undiscovered occurrences, so all metrics are lower bounds. Radiometric data could not be used because of incomplete coverage, and whole-rock geochemistry is too sparse to serve as an input. Eleven gravity and magnetic layers are proprietary Getech products; equivalent derivatives can be computed from the public parent grids, but results will differ somewhat. The provenance of the fault compilation should be confirmed against the original data release before publication. The occurrence-class information (Deposit/Prospect, Showing, Occurrence) was not used for weighting; with 3–8 occurrences per class, confidence-weighted training could not be evaluated reliably. Finally, maps are per-region percentiles and are not calibrated probabilities. At the national scale, 12 of the 66 occurrences lie outside the three study provinces and four of them form one cluster (Ashram), so the "rest of Canada" group is small and spatially dependent; the national labels come from two compilations: a third (Huston, 2024), used by a parallel national study, would add one deposit (Hoidas Lake), and the five records of Orris et al. (2018) whose carbonatite affinity is marked as uncertain, which we excluded, would add five more (one of them Clay-Howells, our external check); and the 0.1° lattice samples high latitudes more densely per unit area.

6 Conclusions

For carbonatite-hosted REE prospectivity mapping in Canada, with 66 occurrences nationally and 12–27 per province:

1. Rebuilt on these data, the national S2-grid design of Parsa et al. (2024) gives a test AUC of 1.000; random splitting of augmented child cells and negatives drawn from the least similar ground each suffice for it. With one label per occurrence, random background and a realistic prevalence, the same forest reaches an AUC of 0.903 and an AP of 0.292.
2. Models do not transfer between geological provinces: a province's model ranks another province's occurrences at or below chance in most cases, the provinces are separable in the evidence layers, and several layers mark occurrences in opposite

directions in different provinces. Prospectivity models for these occurrences should therefore be built per province.

3. In the Superior Province a semi-supervised Deviation Network ranks held-out occurrences better than Random Forest, XGBoost, tuned Random Forest and gradient boosting, bagging-PU and nnPU under random and 50–200 km spatially blocked cross-validation (AP 0.45 and 0.42–0.25 against 0.27–0.28 and 0.20–0.12 for RF and XGBoost), in every repeat except one against XGBoost; with 15 occurrences the margin over RF is consistent but not statistically conclusive. Quantile-normal inputs and background-heavy batches are needed for this result.
4. Geological map classes move every model more than the choice of model. A single mapped alkalic-intrusion class, covering 6 of 15 Superior occurrences and 0.5% of the background, carries over a third of DevNet's blocked AP, and without point-level map classes DevNet and RF lose about half or more of their skill. Prospectivity studies should report results with and without such classes.
5. In the Cordilleran Orogen and the Grenville Province the tree-based learners lead under random splits, a forest given only coordinates matches them in Cordillera and comes close in Grenville, and no model transfers between blocks; the number, independence and geological coherence of the labels, not data coverage, explain the contrast.
6. Synthetic positives from jitter, SMOTE or a tabular WGAN-GP, generated strictly within training folds, gave DevNet no gain and RF only gains that vanish at 200 km.
7. RF and DevNet agree on the direction of all 30 leading drivers across the three provinces. DevNet's Superior map places the untrained Clay-Howells Fe–REE–Nb resource at its 97th percentile and favours greenstone-hosted settings like the dominant Abitibi cluster.

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Declarations

Funding. Not applicable.

Competing interests. The authors declare no competing interests.

Data availability. Occurrence labels are derived from Orris et al. (2018) and Parsa et al. (2024); the deduplicated provincial and national tables and the study-area outlines are distributed with the code. Public layers are available from McCafferty et al. (2023), Jobin et al. (2017), Miles and Oneschuk (2016), D.C. Bradley (2008) and Wheeler et al. (1996). Eleven processed gravity and magnetic grids were supplied by Getech Group plc under licence and cannot be redistributed.

Code availability. The pipeline, every analysis in this paper and the scripts that generate its tables and figures are available at https://github.com/miaozhou-312/ree-prospectivity-paper.

Appendix A Additional results

Table A1 AUC (median over six repeats) for the runs of Table 4. Bold: highest AUC in each row.

Region	Split	DevNet	RF	XGBoost	Bagging-PU	nnPU
Superior	random	0.972	**0.983**	0.976	0.976	0.838
	50 km	**0.973**	0.963	0.966	0.965	0.824
	100 km	**0.950**	0.904	0.922	0.945	0.832
	200 km	0.783	**0.880**	0.826	0.855	0.814
Cordillera	random	0.914	0.931	0.927	**0.936**	0.553
	50 km	0.863	0.849	0.852	**0.866**	0.545
	100 km	0.790	0.776	0.803	**0.847**	0.541
	200 km	**0.821**	0.740	0.783	0.810	0.550
Grenville	random	0.799	0.838	0.857	**0.898**	0.542
	50 km	0.657	0.736	0.709	**0.780**	0.536
	100 km	0.661	0.538	0.621	**0.687**	0.568
	200 km	0.568	0.557	0.584	**0.671**	0.565

Table A2 Sensitivity of nnPU to the assumed class prior π, Superior (random cross-validation; median and 95% interval over six repeats).

π	AUC	AP
0.005	0.811 [0.779, 0.861]	0.133 [0.099, 0.208]
0.01	0.838 [0.794, 0.892]	0.152 [0.125, 0.191]
0.02	0.869 [0.825, 0.916]	0.182 [0.143, 0.263]
0.05	0.902 [0.885, 0.924]	0.264 [0.179, 0.311]

Appendix B Synthetic positives: generators and spatial blocking

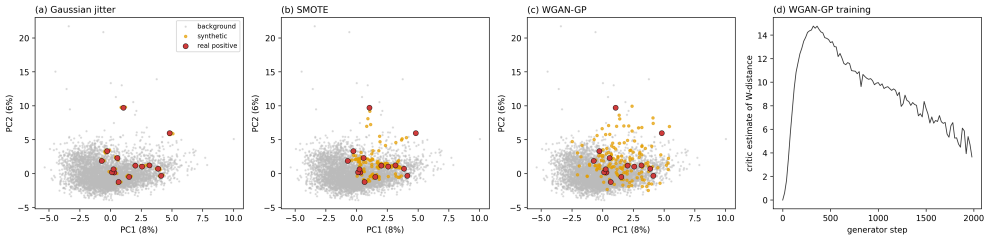


Fig. B1 (a–c) Synthetic positives from each generator (fitted to all 15 Superior positives) in the first two principal components of background and real positives. (d) WGAN-GP critic estimate of the Wasserstein distance during training.

Table B3 Superior average precision with fold-safe synthetic positives under spatially blocked cross-validation (median and 95% interval over six repeats).

Blocks	Model	Augmentation	AP
100 km	DevNet	none	0.343 [0.184, 0.452]
		Gaussian jitter 10×	0.336 [0.163, 0.456]
		SMOTE 3×	0.346 [0.166, 0.448]
		WGAN-GP 3×	0.342 [0.220, 0.541]
		WGAN-GP 10×	0.321 [0.238, 0.395]
100 km	RF	none	0.168 [0.067, 0.181]
		Gaussian jitter 10×	0.157 [0.070, 0.246]
		SMOTE 3×	0.154 [0.059, 0.177]
		WGAN-GP 3×	0.175 [0.093, 0.224]
		WGAN-GP 10×	0.219 [0.091, 0.273]
200 km	DevNet	none	0.252 [0.169, 0.342]
		Gaussian jitter 10×	0.247 [0.167, 0.336]
		SMOTE 3×	0.246 [0.162, 0.365]
		WGAN-GP 3×	0.260 [0.172, 0.334]
		WGAN-GP 10×	0.248 [0.244, 0.317]
200 km	RF	none	0.151 [0.106, 0.158]
		Gaussian jitter 10×	0.100 [0.076, 0.129]
		SMOTE 3×	0.104 [0.055, 0.122]
		WGAN-GP 3×	0.122 [0.089, 0.183]
		WGAN-GP 10×	0.137 [0.093, 0.171]

Appendix C Interpretation in the contrasting provinces

Table C4 As Table 9, Cordillera (55 features).

Feature	DevNet rank	DevNet dir.	RF rank	RF dir.
Distance to regional high-grade metam.	4	↓ (-0.97)	1	↓ (-0.86)
Era: Mesozoic	3	↓ (-0.82)	14	↓ (-0.82)
Bouguer anomaly (BA)	13	↓ (-0.96)	3	↓ (-0.87)
Era: Proterozoic	2	↑ (+0.67)	18	↑ (+0.70)
Age contact (H3 cell)	1	↑ (+0.85)	34	↑ (+0.85)
Moho depth	5	↑ (+0.96)	10	↑ (+0.89)
Passive-margin proximity	32	↑ (+0.73)	2	↑ (+0.78)
Distance to regional age contact	6	↑ (+0.96)	11	↑ (+0.90)
DRTP mean (10 km)	8	↓ (-0.95)	7	↓ (-0.84)
Distance to regional greenstone	30	↑ (+0.93)	4	↑ (+0.20)

Table C5 As Table 9, Grenville (58 features).

Feature	DevNet rank	DevNet dir.	RF rank	RF dir.
Deep-source RTP	1	↑ (+0.97)	2	↑ (+0.73)
Distance to regional age contact	4	↓ (-0.99)	1	↓ (-0.77)
DRTP THD tilt	9	↓ (-0.98)	4	↓ (-0.80)
Distance to regional mafic-ultramafic	15	↑ (+0.98)	3	↑ (+0.45)
Distance to regional sedimentary	11	↓ (-0.98)	5	↓ (-0.91)
BA high-pass 40 km std. dev. (10 km)	3	↑ (+0.98)	8	↑ (+0.85)
DRTP mean (10 km)	7	↑ (+0.97)	14	↑ (+0.94)
Deep gravity-worm proximity	2	↑ (+0.98)	23	↑ (+0.80)
Deep magnetic-worm proximity	6	↑ (+0.98)	25	↑ (+0.80)
Moho depth	20	↓ (-0.96)	6	↓ (-0.68)

Appendix D Ablation of DevNet design choices

DevNet's two design choices (Section 3.3) are examined here with the inputs of the main analysis, by changing one element at a time from a configuration with z-scored inputs and label-balanced batches of 16 + 16 rows. All configurations are evaluated with the folds, protocols and seeds of the main results (Table D6). In Superior, quantile inputs alone raise AP under random splits from 0.358 to 0.424, and background-heavy batches alone to 0.395; together they give 0.445, the highest AP under every protocol except 100 km blocks, where the z-scored, background-heavy variant is level (0.349 against 0.343). Training for longer, by contrast, repeats the dozen positives and pushes the network towards memorising their neighbourhoods: with z-scored inputs, ten times more steps left random-split AP unchanged (0.361 against 0.358) but lowered it under 100 km blocks (0.277 against 0.329). The nested early-stopping configuration chooses the number of steps (100–3,000, cosine schedule over 3,000 steps) inside each outer training fold by a 100 km spatial cross-validation (three inner folds, two inner restarts, background purged from inner validation blocks), so the test fold is never used for the choice. Table D7 gives the sensitivity of the final configuration to the deviation margin: smaller margins gave somewhat higher AP in all three provinces (Superior 0.469 with $a = 2$ against 0.445 with $a = 5$), but we keep the value of Pang et al. (2019) rather than tune it on these folds.

Table D6 Ablation of DevNet's design choices: AP (median over six repeats) under random cross-validation and spatial blocks, on identical folds. Inputs: quantile-normal or z-scored; batch: positive + background rows per step; steps: number of gradient steps (nested ES: chosen by nested 100 km spatial cross-validation). Bold: the final configuration. –: not run.

Region	Inputs	Batch	Steps	random	50 km	100 km	200 km
Superior	quantile	16 + 240	300 (final)	**0.445**	**0.418**	**0.343**	**0.252**
	z-score	16 + 240	300	0.395	0.380	0.349	0.221
	quantile	16 + 16	300	0.424	0.347	0.310	0.239
	z-score	16 + 16	300	0.358	0.332	0.329	0.226
	z-score	16 + 16	3,000	0.361	–	0.277	–
	z-score	16 + 16	nested ES	0.367	0.307	0.262	0.218
Cordillera	quantile	16 + 240	300 (final)	**0.341**	**0.129**	**0.055**	**0.076**
	z-score	16 + 240	300	0.297	0.132	0.078	0.065
	quantile	16 + 16	300	0.301	0.124	0.047	0.074
	z-score	16 + 16	300	0.271	0.154	0.085	0.067
	z-score	16 + 16	3,000	0.228	–	0.060	–
	z-score	16 + 16	nested ES	0.279	0.128	0.068	0.057
Grenville	quantile	16 + 240	300 (final)	**0.125**	**0.031**	**0.007**	**0.005**
	z-score	16 + 240	300	0.072	0.023	0.005	0.004
	quantile	16 + 16	300	0.122	0.030	0.006	0.005
	z-score	16 + 16	300	0.059	0.019	0.006	0.004
	z-score	16 + 16	3,000	0.105	–	0.005	–
	z-score	16 + 16	nested ES	0.088	0.028	0.004	0.004

Table D7 Sensitivity of DevNet to the deviation margin a (AP, median over six repeats, random cross-validation). The main results use $a = 5$.

Region	$a = 2$	$a = 3$	$a = 5$	$a = 8$
Superior	0.469	0.453	0.445	0.436
Cordillera	0.367	0.369	0.341	0.313
Grenville	0.183	0.156	0.125	0.120

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